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Enthalpy of Solution of Carbohydrates Using a Modified Differential Scanning Calorimeter

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A differential scanning calorimeter (DSC) was modified for the determination of enthalpies of solution. The measurements were performed on aqueous solutions of the deoxy- and fluoro-deoxy derivatives of D-glucopyranose (Glu) where the OH group on the C1, C2, C3, and C6 is replaced by H (1HGlu, 2HGlu, 3HGlu, and 6HGlu) and by F (1FGlu, 2FGlu, 3FGlu, and 6FGlu), 4-deoxy 4-fluoro-α-D-glucopyranoside (4FGlu), 1-methoxy-α-D-glucopyranoside (αMeOGlu), 1-phenoxy-α-D-glucopyranoside (αPheOGlu), D-mannopyranose (Man), and 3-methoxy-α-D-glucopyranoside (3MeOGlu) at 15.1, 25.0, 35.0, and 45.1°C. The enthalpies of solution $\Delta_s H^0(T)$ ranged from 1.00 ± 0.25 kJ-mol^{-1} for 6HGlu at 15.1°C to 20.4±1.4 for αPhOGlu at 45.1°C and were in good agreement with literature values for Man, αGlu, αMeOGlu, and 3MeOGlu at 25.0 and 35.0°C and for αMeOMan and 2HGlu at 35.0°C. $\Delta_s H^0(T)$ for the derivatives were then extrapolated up to the melting temperature T_m and compared with their enthalpies of fusion, $\Delta_f H$, also determined from DSC measurements. If the agreement between $\Delta_s H^0(T_m)$ and $\Delta_f H$ was within the 95% confidence level, then it was concluded that intermolecular interactions between the carbohydrate molecules in the liquid phase were the same as between the carbohydrate and water molecules in the solution phase. This agreement was observed for aqueous solutions of Man, αGlu, αMeOGlu, 3HGlu, 3FGlu, and 6FGlu.

KEY WORDS: Aqueous solutions; carbohydrates; differential scanning calorimetry; enthalpy of fusion; enthalpy of solution; solution calorimetry.

1. INTRODUCTION

Knowledge of the solvation properties of carbohydrates in solution is important in understanding their interaction with proteins. Carbohydrates are

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important in many cell recognition processes such as cell differentiation and endocytosis. The thermodynamics of protein–carbohydrate interactions include not only interactions between the ligand and the amino acid residues at the binding site but also changes in the solvation of the ligand and protein as they undergo desolvation from the aqueous state in solution to the bound ligand-protein state. Changes in the thermodynamic parameters for the desolvation process have been determined with computational methods using molecular dynamics simulations and suitable potential energy functions.[1] Experimentally, changes in the desolvation enthalpies of liquids have been determined as the difference between their enthalpies of solution and enthalpies of vaporization.[2]

The desolvation enthalpy of a carbohydrate ligand L can be determined from the enthalpy of solution, $\Delta_s H^0$, which is the total enthalpy change for the following processes

$$\mathrm{L(s)} \rightarrow \mathrm{L(1)} \tag{1a}$$

$$\mathrm{L(1)} \rightarrow \mathrm{L(g)} \tag{1b}$$

$$\mathrm{L(g)} + n\mathrm{H_2O(1)} \rightarrow \mathrm{L(H_2O)_n\ (sol)} \tag{1c}$$

Thus,

$$\Delta_s H^0 = \Delta_f H + \Delta_v H^0 - \Delta_d H^0 \tag{1d}$$

where $\Delta_d H^0$ is the standard enthalpy of desolvation which yields,

$$\Delta_d H^0 = \Delta_f H + \Delta_v H^0 - \Delta_{sol} H^0 \tag{2}$$

The enthalpy of solution and the enthalpy of melting can be determined experimentally with a differential scanning calorimeter (DSC). If $\Delta_s H^0(T) = \Delta_f H(T)$, then this yields the special case where $\Delta_d H^0(T) = \Delta_v H^0(T)$ which would result from equality of the intermolecular interactions between the carbohydrate molecules in the liquid phase and the intermolecular interactions between the carbohydrate molecule and the water molecules in the solution phase. A comparison of $\Delta_s H^0(T)$ with $\Delta_f H(T)$ for the carbohydrate ligands would indicate which carbohydrates exhibit this property.

Isoperibol calorimetry is commonly used to determine enthalpies of solution for solids at ambient temperatures. The limiting detectability of the measurements is about 0.1 mJ[3] so that 50 mg quantities are usually required to obtain accurate solution enthalpies. If a DSC could be modified for enthalpy of solution measurements, then a more accurate value for the enthalpy of solution could be obtained with smaller quantities since the detectability of a DSC is about 0.01 mJ. In this investigation, a method is described and evaluated for the determination of $\Delta_s H^0(T)$ by a simple modification of a DSC. This method is applied to the determination of the enthalpies of solution

of the deoxy- and fluoro-deoxy derivatives of D-glucopyranose (Glu) where the OH group on the C1, C2, C3, and C6 is replaced by H (1HGlu, 2HGlu, 3HGlu, and 6HGlu) and by F (1FGlu, 2FGlu, 3FGlu, and 6FGlu), 4-deoxy 4-fluoro-α-D-glucopyranoside (4FGlu), 1-methoxy-α-D-glucopyranoside (αMeOGlu), 1-phenoxy-α-D-glucopyranoside (αPheOGlu), D-mannopyranose (Man), and 3-methoxy-α-D-glucopyranoside (3MeOGlu) at 15.1, 25.0, 35.0, and 45.1°C in aqueous solutions. Values of $\Delta H^0(T)$ for the carbohydrates are then extrapolated up to the melting temperature T_m and compared with their melting enthalpies, also determined from DSC measurements. If $\Delta_s H^0(T) = \Delta_f H$, then it may be concluded that the intermolecular forces between the carbohydrate in the liquid phase are the same as between the carbohydrate and water molecules in the solution phase. Not only is this important in calculating the thermodynamic changes accompanying carbohydrate–protein interactions but also in identifying the hydration properties of the carbohydrates. Apparent molar volumes and compressibilities,[4] as well as earlier dielectric relaxation measurements,[5] indicate that αGlu, 1MeOGlu, and Man form stable hydration complexes with water molecules because of the presence of the equatorial OH groups. Additional NMR and dielectric relaxation measurements[6] on monosaccharide solutions including αGlu and Man solutions confirmed this hypothesis. This may also be evident in a difference between the intermolecular interactions of the carbohydrate molecules in the liquid phase and the intermolecular interactions between the carbohydrate and water molecules in the solution phase since they exist as hydration complexes resulting in non-equality between $\Delta_s H^0$ and $\Delta_f H$. The formation of hydration complexes may also be affected by the replacement of the OH groups by H or methyl or phenyl groups which would eliminate hydrogen bonding with water or by F which serves only as an acceptor in hydrogen bond formation.

2. EXPERIMENTAL

2.1. Materials

The carbohydrate derivatives were reagent grade from Sigma Chemical[2] and dried over Drierite overnight before use. Purities of Man and αMeOMan were mass fraction 0.99 in the α form. The Chemical Abstracts Service

[2]Certain commercial equipment, instruments, and materials are identified in this paper in order to specify the experimental procedure as completely as possible. In no case does this identification imply a recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the material, instrument, or equipment identified is necessarily the best available for the purpose.

registry numbers of the carbohydrates are as follows: Man, 3458-28-4; αMeO-Man, 617-04-9; αGlu, 50-99-7; αMeOGlu, 97-30-3; αPhOGlu, 4630-62-0; 2HGlu, 154-17-6; 2FGlu, 86783-82-6; 3HGlu, 2490-91-7; 3FGlu, 14049-03-7; 3MeOGlu, 13224-94-7; 4FGlu, 62182-11-0; 6HGlu, 7658-08-4; 6FGlu, 34168-77-9. The molar masses in g-mol^{-1} were 180.2 for Man and αGlu, 194.2 for the methoxy derivatives, 164.2 for the deoxy Glu derivatives, 182.1 for the fluoro Glu derivatives, and 256.4 for αPhOGlu. The sodium chloride, dimethylglutaric acid, sodium hydroxide, potassium chloride, manganese chloride, and calcium chloride for preparation of the buffer were reagent quality and also obtained from Sigma Chemical. The 99.99 mass % pure indium sample was kindly provided by Prof. Eric J. Cotts of Binghamton Univeristy. The buffer used for the solution enthalpy measurements was 0.01*M* dimethylglutaric acid adjusted to pH 6.95 ± 0.05 with sodium hydroxide and containing 1 m*M* each of $MnCl_2$ and $CaCl_2$ and 0.15*M* sodium chloride. This was the solvent used in earlier lectin binding studies.[7] At 25.0°C, the enthalpy of solution measurements were also performed with distilled water to determine if the presence of the additional solutes in the buffered solutions had any effect on $\Delta_s H^0$. Some of the Man and αGlu solutions were analyzed by reacting an aliquot of the solution with *o*-toluidine reagent in trichloroacetic acid at 100°C (Sigma Diagnostic Kit 635) as described previously.[7] The concentrations thus determined agreed to within 3% with the concentrations determined from weighings.

2.2. Enthalpy of Solution Determinations

The enthalpies of solution were determined from measurements with a modified Hart 7707 DSC operated in the isothermal mode. The modification consisted of replacing the two adiabatic enclosure covers above each vessel holder with enclosure covers with 2 mm diameter holes through their centers. The four vessels, one reference vessel and three solution vessels, containing about 0.5 g of the buffer solution were placed in their vessel holders uncapped. Then the enclosure covers were placed over the vessel holders and small pyrex funnels having ODs ranging from 2 mm to 7 mm and about 3 cm long were inserted through the enclosure cover holes so that the stem ends of the funnels were positioned above the surface of the buffer solution contained in each of the three solution vessels. The minimum operating temperature was 15.0°C to avoid the possibility of condensation from room air at lower temperatures in the adiabatic enclosure. Temperatures greater than 46°C were avoided to minimize evaporation of the solvent. The DSC was allowed to equilibrate isothermally until a constant baseline reading was obtained; this took about 45 minutes. Then, a known mass of carbohydrate was poured into the funnel which had already been weighed and the DSC response was

recorded. This usually resulted in an endothermic peak of the power output versus time as shown in Fig. 1. Lack of stirring to insure a uniform concentration was partially compensated by a agitation of the solution caused by pouring the powdered sample into the small 0.5 mL volume of solvent. Complete solubilization of the solute was evident by the return of the signal to the original baseline about 15 min after pouring in the solute. The scan was terminated about 30 min after the peak returned to the baseline. After the measurements, the funnel was reweighted to determine if any of the carbohydrate had adhered to the sides of the funnel. The mass of carbohydrate was accordingly corrected for this. The area of the peak was determined using the EXAM program[8] and a straight baseline. The enthalpy of dropping the sample into the DSC vessel was determined by dropping samples of the same carbohydrate derivative into the empty DSC vessel. It was found that this contribution on a molar basis was the same for all the carbohydrates and it increased from -1.61 ± 0.09 kJ-mol^{-1} at 15.1°C to -0.21 ± 0.09 at 25.0°C, to 1.53 ± 0.14 at 35.0°C, and to 2.62 ± 0.24 at 45.1°C. This contribution changes from a negative to a positive value as the temperature goes from below to above room temperature, indicating that the source of the change is at room temperature and absorbs heat from higher temperature surroundings as well as releases heat to lower temperature surroundings. Since these contributions were the same and change from a negative to a positive contribution in going from 15.1 to 35.0°C, they are the result of heat exchanged between the solid sample equilibrated at room temperature (24°C) and the DSC vessel equilibrated at a different temperature. Values for $\Delta_s H^0$ were determined by

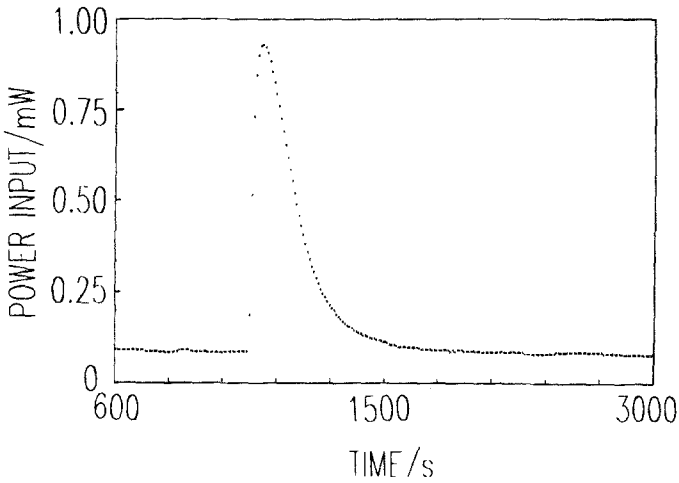

Fig. 1. An isothermal DSC scan of the dissolution of 2.56 mg 4FGlu in DMG buffer at 45.9°C.

subtracting these values from the uncorrected $\Delta_s H^0$ values. The calibration constants determined at 15.1, 25.0, 35.0°C, and 45.1°C with the internal heaters in this modified operating set-up were found to be within 1% of the calibration constants determined in the normal DSC operating mode with the original enclosure covers. A $\Delta_s H^0$ of 16.9 ± 0.5 kJ-mol^{-1} in water after correcting for the dropping enthalpy was determined for KCl at 25.0°C in agreement with the enthalpy of solution of 17.47±0.07 for 1 mole of KCl in 2000 moles of H_2O.[9] The errors are the estimated uncertainties as described below. To determine if the method exhibited any temperature dependence, the $\Delta_s H^0$ for KCl was also determined at 45.1°C. The corrected value of 13.6±0.2 determined for KCl at this temperature was in agreement at the 95% confidence level with the value of 14.1 ±0.1 calculated from the literature values of $\Delta_s H$ (KCl) at 25.0°C and $\Delta_s C_p$ of −166 J-mol^{-1}-K^{-1}.[10]

A Microcal Omega titration calorimeter was used to measure the enthalpies of dilution of the carbohydrate solutions in the final concentration range of the dissolved carbohydrates. Ten μL aliquots of the carbohydrate solutions were injected into the buffer solution in the 1.38 mL vessel of the titration calorimeter and any heat exchanged with the reference vessel was recorded. Dilution of up to 100 m*M* solutions of the carbohydrate by this factor of 1380/10 = 138 did not exhibit any heat exchange although the sensitivity was about 1 μJ.

2.3. Enthalpy of Melting Determinations

The enthalpies of melting were determined with a Perkin–Elmer DSC-2 for all the derivatives which melt above the maximum operating temperature of the Hart DSC (115°C). Only $\Delta_f H$ for 3FGlu which melts below 110°C was determined with the Hart DSC. Milligram quantities of the carbohydrate were sealed in 0.1 mL aluminum capsules for the $\Delta_f H$ determinations to minimize exposure of the carbohydrate to air and consequent oxidation upon melting. For measurements with the Hart DSC operated at a scan rate of 8.33×10^{-4} K-s^{-1}, the sealed aluminum capsules were placed inside the DSC vessels. The Perkin–Elmer DSC was operated at a scan rate of 0.167 K-s^{-1} up to 230°C and the heat capacity changes were recorded every 0.5 K. Areas under the melting transition peaks were determined with the EXAM Program[8] using a straight baseline. The calibration of the Perkin–Elmer DSC was performed with an indium sample which melts at 154°C with a $\Delta_f H$ = 3.26 kJ-mol^{-1}.[11] The melting temperature was determined as the temperature where the tangent to the rising portion of the melting peak intersects with a straight baseline extrapolated between the pre- and post-transitional baselines.

2.4. Calculated Enthalpy of Solution at the Melting Point

The enthalpy of solution at the melting temperature was calculated with the following equation,

$$\Delta_s H^0(T_m) = \Delta_s H^0(318.3) + \Delta_s C_p(T_m - 318.3) \quad (3)$$

which assumes a constant value for the heat capacity change accompanying dissolution, $\Delta_s C_p$, over the temperature range from 15.1°C up to T_m. Values for $\Delta_s C_p$ are the slopes of linear least squares fits of ΔH_s at 15.1, 25.0, 35.0, and 45.1°C to the temperature.

2.5. Error Analysis

Values for $\Delta_s H^0$ at each temperature were determined from an average of at least three DSC measurements for each carbohydrate over a final concentration range from 5 to 44 mM. To determine how the known sources of error in the method affect the determination of $\Delta_s H^0$, the standard deviation for each mean value of $\Delta_s H^0$ was compared with the combined standard uncertainty in the determination of $\Delta_s H^0$. The combined standard uncertainty in $\Delta_s H^0$ can be estimated from possible errors in the mass m of the sample dropped into the solvent, in the determination of the DSC calibration constant C for converting the DSC voltage readings into power readings, and in the determination of the peak areas A from solution of the sample in water and from the enthalpy of dropping the sample into the DSC (δ). There may also be contributions from incomplete dissolution of the sample due to the lack of stirring. Since this contribution would be expected to increase with the amount of sample and it was observed that $\Delta_s H^0$ is independent of sample size, this contribution is expected to be negligible. Since

$$\Delta_s H^0 = MCA/m - \delta \quad (4)$$

where M is the molar mass, the combined standard uncertainty in $\Delta_s H^0$ is

$$\sigma(\Delta_s H^0) = M\{[A/m\ \sigma(C)]^2 + (C/m\ \sigma(A))^2 + (CA/m^2\ \sigma(m))^2 + \sigma(\delta)/M)^2\}^{1/2} \quad (5)$$

The relative uncertainty in the determination of C is about 0.002 over the temperature range from 15 to 46°C. With a straight baseline in alignment with the pre- and post-peak baselines, the relative uncertainty in the area resulting mainly from variation of the integration limits is about 0.02. The maximum relative uncertainty in the mass, determined to ±0.02 mg with a Mettler AE163, varies from 0.04 for a 1 mg sample to 0.01 for a 4 mg sample. The uncertainty in the measured enthalpy of the dropping effect is estimated to be from $0.01\Delta_s H^0$ to $0.02\Delta_s H^0$. Therefore, the combined standard uncertainty from Eq (5) is about 0.04 $\Delta_s H^0$ for an average 2.5 mg sample.

Values of $\Delta_f H$ given in Table II are the averages of the results of DSC scans on two different samples. Since the mass of the sample was determined with a more accurate hanging pan balance, the principle sources of error are a relative uncertainty of 0.01 in the melting peak area determined by the EXAM program[8] and 0.02 relative uncertainty in the calibration factor determined by several meltings of the indium. The estimated combined standard uncertainty is about 0.02 $\Delta_f H$.

Uncertainties in the calculated enthalpies of solution at T_m were calculated using

$$\sigma(\Delta_s H^0(T_m)) = [\sigma[\Delta_s H^0(45.2°C)]^2 + \sigma(\Delta_s C_p)^2(T_m - 45.2)^2]^{1/2} \quad (6)$$

where $\sigma(\Delta_s H^0(45.2°C)$ and $\sigma(\Delta_s C_p)$ are the standard deviations of the mean values of, respectively, $\Delta_s H^0$ at 45.2°C and $\Delta_s C_p$.

3. RESULTS

The endothermic peak shown in Fig. 1 for the dissolution of 2.56 mg 4FGlu in DMG buffer at 45.9°C is typical for dissolution of a Glu derivative. Rapid dissolution of the Glu derivative is important since conversion from the α form to the β form could occur during dissolution. For example, 6% of αGlu converts to the β form in 600 s in solution at pH 7.0 and 25°C.[12] The decay following the peak maximum arises from the slow response time of 138 s for the DSC and slow dissolution rate of the carbohydrate. Correction for the DSC response time by the Tian equation in the EXAM Program reduces the observed peak decay time from 144 s to 30 s which can be attributed to the time required for dissolution of the sample to go to 63% completion. For αGlu, less than 1% of the initial α form would be converted to the β form during the time it takes for 63% of the Glu to dissolve and only 6% during 10 min observed for the peak to return to a constant baseline. Continuation of the slow α to β conversion after return to the baseline was not apparent since the baseline remained constant, indicating that the enthalpy contribution from this conversion was within the baseline noise. Stirring of the solution would accelerate dissolution of the sample but would also add noise to the DSC baseline.[3]

Values for $\Delta_s H^0$ are the average values of at least three determinations over the final concentration range from 5 to 44 m*M*. $\Delta_s H^0$ does not exhibit any dependency on the concentration over this range. Since no enthalpy of dilution was observed with the titration calorimeter over this concentration range, values of $\Delta_s H^0$ determined in this investigation are considered to be the same as for enthalpies of solution at infinite dilution.

Determinations of $\Delta_s H^0$ for the carbohydrates in water at 25.0°C and in DMG buffer at 15.1°C, 25.0°C, 35.0°C, and 45.1°C are presented in Table I. The errors in Table I are the single standard deviations of the average $\Delta_s H^0$ determination. As shown in Table I, values of $\Delta_s H^0$ are within two standard deviations the same in both water and the DMG buffer. They range from 0.98 kJ-mol^{-1} for 6HGlu to 15.2 kJ-mol^{-1} for 2FGlu at 25.0°C, indicating that substituents on the glucopyranoside ring can substantially affect the enthalpy of solution. The values at 25.0 and 35.0°C are compared with literature values (13–15) in Table I. For Man, αGlu and 3MeOGlu at 25.0°C and 35.0°C and for αMeOMan, αMeOGlu, and 2HGlu at 35.0°C, the agreement with the literature values for $\Delta_s H^0$ is within experimental error while the agreement for αMeOMan, αMeOMan and 2HGlu at 25.0°C is poor. The less than 15% difference between the literature and experimental values of $\Delta_s H^0$ for αMeOMan, αMeOGlu, and 2HGlu at 25.0°C is the same as between the literature values of Jasra and Ahluwalia[13] and Balk and Somsen[14] for the $\Delta_s H^0$(25°C) of Man.

The average of the standard deviations for all the $\Delta_s H^0$ determinations, except those values for 6HGlu in Table I is 3.7%, well within the uncertainty of 4.0% estimated from Eq. (5). There appears to be a slight temperature dependence in the standard deviations since the average of standard deviation at 25.0°C is 2.2% and this average increases to 3.4% at 15.1°C, 4.2% at 35.0°C, and 5.1% at 45.1°C. Apparently, all the sources of error in the method have been accounted for in Eq. (5).

Over the temperature range 15 to 45°C, $\Delta_s H^0$ varies linearly with temperature as shown in Fig. 2 for αGlu. The value of $\Delta_s C_p$ from the slope of the plot in Fig. 2 is 0.160 ± 0.011 kJ-mol^{-1}-K^{-1} in fair agreement with the literature value of 0.121 at 25.0°C[16] obtained from the difference between the heat capacity of solid Glu, 0.223 kJ-mol^{-1} K^{-1},[17] and the solution value of 0.344[16] Values of $\Delta_s C_p$ are also presented in Table I. For Man, αMeOMan, αMeOGlu, and 3MeOGlu the values are within the literature values of Jasra and Ahluwalia.[13] The heat capacity changes can be classified into the following two categories: (1) values for αMeOMan, αMeOGlu, and αPhOGlu are ≥ 0.2 and (2) values for the other carbohydrates in Table I are ~0.1. The large values of the alkoxy and phenoxy substituted derivatives in category (1) can be rationalized in terms of the structure promoting properties of the alkoxy and phenoxy groups in water. Comparison of the category 2 values with an average heat capacity change of 0.200 for the solid state of the derivatives[13] show that the interaction energy of these derivatives with water is close to that in the crystal lattice.

The $T_m S$ for the Glu derivatives range from 105°C for 3FGlu to 182°C for αMeOMan and are presented in Table II along with the enthalpies of melting. The enthalpy of melting for 1FGlu was not determined since its

Table I. Enthalpies of Solutions of Pyranoside Derivatives in DMG Buffer and in Water and their Solution Heat Capacity Changes in DMG Buffer[a]

Derivative	$\Delta_s H^0$ (kJ-mol^{-1}) 15.1°C	25.0°C	35.0°C	45.1°C	$\Delta_s C_p^e$
Man	5.92±0.28	8.02±0.09 (6.86±0.06) (7.87[b]) 8.08±0.48 in water	8.83±0.42 (8.34±0.09)	9.88±0.93	0.134±0.011 (0.148±0.011)
αMeOMan	7.31±0.11	10.5±0.1 (9.09±0.07) 10.0±0.2 in water	11.7±0.4 (11.16±0.06)	16.2±0.8	0.293±0.010 (0.207±0.009)
αGlu	8.95±0.37	11.0±0.2 (10.7±0.04) 11.0±0.2 in water (11.1±0.3[c])	12.2±0.5 (12.04±0.12)	13.9±0.8	0.160±0.011 (0.134±0.012) (0.124[d])
αMeOGlu	2.16±0.10	4.55±0.22 (3.67±0.07) 4.32±0.11 in water	6.39±0.38 (5.65±0.03)	8.88±0.64	0.219±0.009 (0.200±0.010)
αPhOGlu	8.53±0.25	13.0±0.4	14.0±1.1	18.7±1.4	0.314±0.050
1HGlu	7.24±0.39	8.81±0.17	9.37±0.24	10.6±0.4	0.111±0.004
1FGlu	9.36±0.13	10.6±0.6	10.3±0.2	11.9±0.3	0.082±0.015
2HGlu	9.87±0.17	10.5±0.1 (9.43±0.06) 10.3±0.1 in water	10.2±0.5 (11.23±0.15)	12.3±0.3	0.082±0.006 (0.180±0.016)
2FGlu	14.2±0.2	15.2±0.2 14.4±0.4 in water	14.9±0.7	17.0±0.6	0.093±0.003
3HGlu	9.84±0.20	13.7±0.1	12.9±0.2	15.7±0.3	0.189±0.030
3FGlu	7.97±0.22	10.5±0.1 10.4±0.2 in water	9.97±0.24	11.4±0.3	0.112±0.010
3MeOGlu	5.17±0.26	8.16±0.10 (8.27±0.07)	8.97±0.71 (10.19±0.05)	10.8±0.7	0.189±0.002 (0.192±0.008)
4FGlu	12.9±0.1	12.6±0.5	14.5±0.4	16.5±1.0	0.194±0.002
6HGlu	0.00±0.12	0.98±0.10	1.40±0.17	2.96±0.42	0.099±0.001
6FGlu	8.45±0.12	9.36±0.20	8.97±0.33	13.2±0.9	0.163±.024

[a]Literature value are in parentheses. The errors are the standard deviations of at least three enthalpy determinations and do not include systematic errors from uncertainties in the number of moles of each substance. Unless otherwise noted all literature values are from Ref. 13; [b]From Ref. 14; [c]From Ref. 15; [d]From Ref. 16; [e]Units: J-mol^{-1}-K^{-1}.

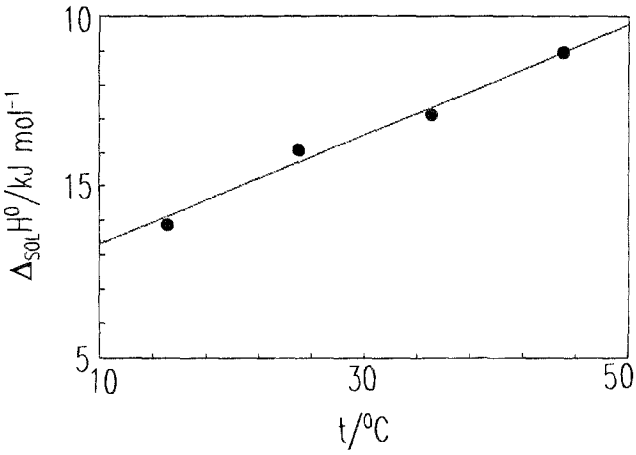

Fig. 2. The dependence of $\Delta_s H^0$ for αGlu on temperature. The straight line is the best least squares fit to the data.

DSC scan exhibits only an apparent phase transition around 90°C similar to that observed for amorphous fructose.[18] The shapes of the melting peaks of the other derivatives appear to reflect a high degree of purity with the exception of 4FGlu which is broad and may reflect decomposition of the derivative upon melting. A literature value of 31.4 kJ-mol^{-1}[19] for αGlu is close to the

Table II. Comparison of the Estimated Enthalpy of Solution at the Melting Temperature (Est. $\Delta_s H^0(T_m)$) with the Enthalpy of Melting ($\Delta_f H$) for the Glucopyranoside Derivatives[a]

Derivative	Melting temp (°C)	$\Delta_f H$	Est. $\Delta_s H^0(T_m)$	Est. $\Delta_s H^0(T_m)/\Delta_m H$
Man	118.0±1.0	24.7±0.7	19.6±1.2	0.80±0.05
αMeOMan	182.0±1.0	44.7±1.0	56.3±0.9	1.26±0.03
αGlu	150.0±0.5	34.3±0.3	30.7±3.4	0.90±0.10
αMeOGlu	151.0±1.0	37.6±0.6	32.1±1.2	0.85±0.0.4
αPhOGlu	156.0±0.5	39.0±1.0	53.5±5.0	1.37±0.14
1HGlu	130.0±0.5	27.4±0.2	20.0±0.5	0.73±0.02
2HGlu	125.5±1.0	34.5±0.9	18.9±1.1	0.55±0.04
2FGlu	154.0±0.5	38.2±0.1	27.1±0.6	0.71±0.02
3HGlu	114.0±0.5	32.6±0.2	28.7±2.3	0.88±0.07
3FGlu	105.0±0.1	18.3±0.6	18.1±0.8	0.99±0.06
3MeOGlu	152.0±0.5	41.3±1.8	31.0±0.7	0.75±0.03
6HGlu	136.0±0.5	22.7±0.5	13.7±0.5	0.59±0.02
6FGlu	139.0±0.2	27.2±0.2	28.5±2.4	1.08±0.11

[a]The errors are the standard deviations of at least three enthalpy determinations and do not include systematic errors from uncertainties in the number of moles of each substance.

value of 34.3±0.3 given in Table II. With the exception of 3MeOGlu, the average of the standard deviations of the 13 $\Delta_f H$ determinations is 0.019 $\Delta_f H$, within the estimated uncertainty of $0.02\Delta_f H$.

4. DISCUSSION

Estimations of $\Delta_s H^0(T_m)$ using Eq. (3) are presented in Table II. The agreement between these estimated values and $\Delta_f H$ close to the 95% confidence level, where the values are almost within two standard deviations, is observed for Man, αGlu, αMeOGlu, 3HGlu, 3FGlu, and 6FGlu. Thus, the intermolecular interactions between these derivatives in the liquid phase are the same as between the derivative and water molecules in the solution phase. This implies that they approach ideal behavior in solution since from the van't Hoff equation,

$$\Delta_s H^0 = RT^2\, d[\ln(x)]/dT \tag{7}$$

where $R = 8.315$ J-mol^{-1}-K^{-1} and x is the mole fraction at saturation and since $\Delta_s H^0(T_m) = \Delta_f H$, then

$$\Delta_f H = RT^2\, d[\ln(x)]/dT \tag{8}$$

which yields upon integration,

$$\ln(x) = -\Delta_f H/R(1/T - 1/T_m) \tag{9}$$

the ideal law of solubility.[20]

For αMeOMan and αPhOGlu, the estimated enthalpies of solution at T_m are greater than $\Delta_f H$ indicating that more than the enthalpy of melting is required to solubilize the derivative. For 1HGlu, 2HGlu, 2FGlu, 3MeOGlu, and 6HGlu, the estimated $\Delta_s H^0(T_m)$ values are less than the corresponding $\Delta_f H$ values, indicating that heat is released upon solubilization of these Glu derivatives. Thus, solutions of αMeOMan, αPhOGlu, 1HGlu, 2HGlu, 2FGlu, 3MeOGlu, and 6HGlu approach non-ideal behavior in water. This difference in the solution behavior of the carbohydrates cannot be accounted for by differences in their conformations in water since replacement of an OH group by H, F, or a bulky one such as methoxy has little effect on its conformation.[21] Dielectric relaxation measurements[5,6] of aqueous solutions of Man, αGlu, and αMeOGlu indicate that they form hydrates with water below 70°C. Only by considering them as hydrate complexes in solution, do their solutions approach ideal behavior.[6] However, the formation of hydrates for these carbohydrates is not evident in this investigation since the enthalpy of solution is the same as the enthalpy of melting. Perhaps the assumption that $\Delta_s C_p$ is constant with temperature over the temperature range from 45 to 150°C is incorrect. Very precise measurements[16] on one component of $\Delta_s C_p$, the heat

capacity of the aqueous phase, $C_{p,2}$, of αGlu solutions from 25.0 to 53.9°C, for example, exhibit an increase with temperature of about .001 kJ-K^{-2}-mol^{-1} which may be compensated by a temperature dependence of the other component of $C_{p,s}$, the heat capacity of the solid phase.

The method of measuring the enthalpy of solution with a modified DSC yields results with a 4% maximum estimated uncertainty. It can be applied to other DSC's including those with fixed vessels. Necessary criteria are that they operate in the range suitable for measuring large enthalpies of solution, (*e.g.*, a 1 micromole sample with $\Delta_s H^0$ = 30 kJ-mol^{-1} will absorb 30 mJ of heat) and that the sample can be expeditiously introduced into the DSC vessel. This method has some advantages over the usual isoperibol calorimeters used for determining enthalpies of solution. The principal advantage is that it requires smaller amounts of sample since the detectability is more than an order of magnitude greater than that for an isoperibol calorimeter. The solvent equilibration times are less because of the smaller volumes (1 mL) used. The detectability for a solution calorimeter is limited to 0.1 mJ[3] by the noise introduced in the breaking of a glass ampoule containing the sample and in the stirring of the solution. Error introduced into the determinations by the "dropping effect" may be reduced by equilibrating the solid at the temperature of the DSC vessel prior to pouring it into the solvent.

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